

cloud computing and data centre hub, adds Sadasiva.

In addition, the roll-out and deployment of 5G, IoT, smart cities and the advent of edge computing will drive data centre demand in the future. No wonder then, that AdaniConneX, in addition to developing full-scale data centres, says it will also develop a portfolio of edge data centres strategically located throughout India. This will support the need for more proximate capacity. These edge sites are designed and planned to scale with demand and become full-scale data centre campuses easily. While there isn't a single model that fits all, edge data centres are said to be crucial for certain sectors that are dependent on low latency. It will help with real-time information monitoring and processing,

technology and infrastructure. The recurring costs in a data centre include a range of things such as hardware maintenance and upgrade, upkeep of the facility and a skilled workforce."

Given that the need for data centres is set to go through the roof in the coming years, there is enough space for more players to grab a piece of the growing pie. From just a single-digit count of data centre players in India, there has been an influx of many players just in the last three-four years. However, Sharma of NTT, who has over 25 years of experience in this space, says, "Data centre is a very capex-heavy industry. So, definitely, not anybody or everybody can afford it. But yes, these companies [who survive] are the ones who have the strength,



BACKUP/DISASTER RECOVERY

Data centres have backup and redundant systems to ensure that data is not lost in case of a system failure or any other disaster



SECURITY MEASURES

As data centres contain sensitive information and valuable equipment, they have strict security measures to prevent unauthorised access. This includes physical as well as cyber-security measures

TYPES OF SOLUTIONS

1 Captive in-house data centre: Having a data centre within enterprise premises

2 Hyperscale data centre: Primarily catering to hyperscalers like Google, Microsoft and Amazon. These are standalone facilities built to suit such a customer or purpose-built for customers that take up minimum capacities of 4-5 MW each

3 Wholesale co-location: Caters to requirements of businesses and enterprises in the range upwards of approximately 500kW per customer

4 Retail co-location: This caters to any size, even a single rack of 6/7 kW

| 59

management of expanding IoT applications and delivery of improved service provider experiences. Edge computing will transform how businesses leverage and benefit from data. As processing is done close to the data source, it allows for reduced response time while producing precise and desired outcomes.

However, there are a few challenges to India's fledgeling data centre industry. Though approval processes have improved quite significantly after the industry was granted infra status, it still takes a couple of months at times. Data centres demand huge investments, but the return on investment is very slow, which might act as a deterrent for some. Says Gupta: "The costs involved in setting up a data centre or a data centre park can vary depending on factors such as the location, size, design,

muscle power and finances to run it. A lot of them will test the waters in the coming years." And this trend is expected to continue in the coming years with healthy competition in the space. Ultimately, just like the telecom industry, some of them will exist, while some of them will sell out, consolidating the industry into five or six major players. For instance, Equinix acquired GPX India in an all-cash transaction valued at \$161 million in 2020 to enter the Indian data centre market.

All said and done, as businesses embrace local service providers, data protection and localisation, be it a handful of colo players or a full stack operator, data centres in India is a greenfield space with strong odds. **BT**

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